

Proposed proposed to SSMJ

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Matrix Equation

Statement of the Problem: Find the 2×2 matrices M with real entries that satisfy the equation:

$$M^2 - 5M = \begin{pmatrix} 0 & 9 \\ 4 & 0 \end{pmatrix}$$

Solution of the Problem:

Notice that

$$M^2 - 5M + 6I = \begin{pmatrix} 6 & 9 \\ 4 & 6 \end{pmatrix}$$

$$(M - 2I)(M - 3I) = \begin{pmatrix} 6 & 9 \\ 4 & 6 \end{pmatrix}$$

Taking determinant on both sides

$$(|M - 2I|) (|M - 3I|) = 0$$

$$|M - 2I| = 0 \text{ or } |M - 3I| = 0$$

This means, if λ_1 be one of the eigenvalues of the matrix M , then

$$\lambda_1 = 2 \text{ or } 3 \tag{1}$$

Similarly, also notice that

$$M^2 - 5M - 6I = \begin{pmatrix} -6 & 9 \\ 4 & -6 \end{pmatrix}$$

$$(M + I)(M - 6I) = \begin{pmatrix} -6 & 9 \\ 4 & -6 \end{pmatrix}$$

Taking determinant on both sides

$$(|M + I|)(|M - 6I|) = 0$$

$$|M + I| = 0 \text{ or } |M - 6I| = 0$$

This means, if λ_2 be the other eigenvalue of the matrix M, then

$$\lambda_2 = -1 \text{ or } 6 \quad (2)$$

We won't explore any further for eigenvalues because a 2×2 Matrix can only have two eigenvalues.

We have four possible pairs of eigenvalues for matrix M using equation (1) and (2).

$$(\lambda_1, \lambda_2) = (2, -1) \text{ or } (2, 6) \text{ or } (3, -1) \text{ or } (3, 6)$$

We know, for a 2×2 Matrix, the characteristic equation is :

$$\lambda^2 - \text{Tr}(M)\lambda + \text{Det}(M) = 0$$

where $\text{Tr}(M)$ stands for Trace of a Matrix and $\text{Det}(M)$ stands for Determinant of a Matrix.

By Cayley-Hamilton theorem, any matrix M satisfies its characteristic equation. Thus,

$$M^2 - \text{Tr}(M)M + \text{Det}(M) = 0$$

Using the fact that $\text{Tr}(M) = \lambda_1 + \lambda_2$ and $\text{Det}(M) = \lambda_1 \lambda_2$,

For eigenvalue pair (2,-1), we get

$$M^2 - M - 2I = 0 \implies M^2 = M + 2I \quad (3)$$

which is a necessary but not sufficient equation. Using the Problem statement and applying equation (3) on it

$$\begin{aligned} M^2 - 5M &= \begin{pmatrix} 0 & 9 \\ 4 & 0 \end{pmatrix} \\ M + 2I - 5M &= \begin{pmatrix} 0 & 9 \\ 4 & 0 \end{pmatrix} \implies -4M + 2I = \begin{pmatrix} 0 & 9 \\ 4 & 0 \end{pmatrix} \\ \implies -4M &= \begin{pmatrix} -2 & 9 \\ 4 & -2 \end{pmatrix} \implies M = \begin{pmatrix} \frac{1}{2} & -\frac{9}{4} \\ -1 & \frac{1}{2} \end{pmatrix} \end{aligned}$$

Repeating the same procedure, for eigenvalue pair (2,6), we get $M = \begin{pmatrix} 4 & 3 \\ \frac{4}{3} & 4 \end{pmatrix}$.

For eigenvalue pair (3,-1), we get $M = \begin{pmatrix} 1 & -3 \\ -\frac{4}{3} & 1 \end{pmatrix}$. For eigenvalue pair (3,6),

we get $M = \begin{pmatrix} \frac{9}{2} & \frac{9}{4} \\ 1 & \frac{9}{2} \end{pmatrix}$. These four solutions are all possible solutions for real matrices M.